

FxP-QNet: A Post-Training Quantizer for the Design of Mixed Low-Precision DNNs with Dynamic Fixed-Point Representation

AHMAD SHAWAHNA¹, SADIQ M. SAIT^{1, 3}, (Senior Member, IEEE), AIMAN EL-MALEH^{1, 3}, (Member, IEEE), AND IRFAN AHMAD^{2, 3}

¹Department of Computer Engineering, King Fahd University of Petroleum & Minerals Dhahran-31261, Saudi Arabia

²Information and Computer Science Department, King Fahd University of Petroleum & Minerals Dhahran-31261, Saudi Arabia

³Interdisciplinary Research Center for Intelligent Secure Systems, King Fahd University of Petroleum & Minerals Dhahran-31261, Saudi Arabia

Corresponding author: Ahmad Shawahna (e-mail: g201206920@kfupm.edu.sa).

This work was supported by the King Fahd University of Petroleum & Minerals Dhahran, Saudi Arabia.

ABSTRACT Deep neural networks (DNNs) have demonstrated their effectiveness in a wide range of computer vision tasks, with the state-of-the-art results obtained through complex and deep structures that require intensive computation and memory. In the past, graphic processing units enabled these breakthroughs because of their greater computational speed. Now-a-days, efficient model inference is crucial for consumer applications on resource-constrained platforms. As a result, there is much interest in the research and development of dedicated deep learning (DL) hardware to improve the throughput and energy efficiency of DNNs. Low-precision representation of DNN data-structures through quantization would bring great benefits to specialized DL hardware especially when expensive floating-point operations can be avoided and replaced by more efficient fixed-point operations. However, the rigorous quantization leads to a severe accuracy drop. As such, quantization opens a large hyper-parameter space at bit-precision levels, the exploration of which is a major challenge. In this paper, we propose a novel framework referred to as the **Fixed-Point Quantizer of deep neural Networks (FxP-QNet)** that flexibly designs a mixed low-precision DNN for integer-arithmetic-only deployment. Specifically, the **FxP-QNet** gradually adapts the quantization level for each data-structure of each layer based on the trade-off between the network accuracy and the low-precision requirements. Additionally, it employs post-training self-distillation and network prediction error statistics to optimize the quantization of floating-point values into fixed-point numbers. Examining **FxP-QNet**¹ on state-of-the-art architectures and the benchmark ImageNet dataset, we empirically demonstrate the effectiveness of **FxP-QNet** in achieving the accuracy-compression trade-off without the need for training. The results show that **FxP-QNet**-quantized AlexNet, VGG-16, and ResNet-18 reduce the overall memory requirements of their full-precision counterparts by 7.16×, 10.36×, and 6.44× with less than 0.95%, 0.95%, and 1.99% accuracy drop, respectively.

INDEX TERMS Neural Networks, Model Compression, Deep Learning, Quantization, Fixed-Point Arithmetic, Mixed-Precision, Acceleration, Accuracy, Efficient Inference, Resource-Constrained Devices.

I. INTRODUCTION

DEEP neural networks (DNNs) have achieved state-of-the-art results on computer vision tasks, such as image classification [1]–[7], object detection [8], [9], face recognition [10], and semantic segmentation [11]–[13]. Examples of recent fields of DNN applications include robot navigation,

industrial automation, automated driving, aerospace and defense [14]–[16]. These applications heavily depend on the complex DNN structures as well as on the deep and complicated models that over-fit the distribution of numerous training data. However, this also leads to over-parameterization resulting in a dramatic increase in memory and computational requirements for deployment.

A typical DNN often takes hundreds of MB of memory space. For instance, 170MB for ResNet-101 [3], 240MB for AlexNet [6], and 550MB for VGG-16 [7], and requires

The associate editor coordinating the review of this manuscript and approving it for publication was Paolo Crippa.

¹The code of the proposed framework will be available online in <https://github.com/Ahmad-Shawahna/FxP-QNet>.